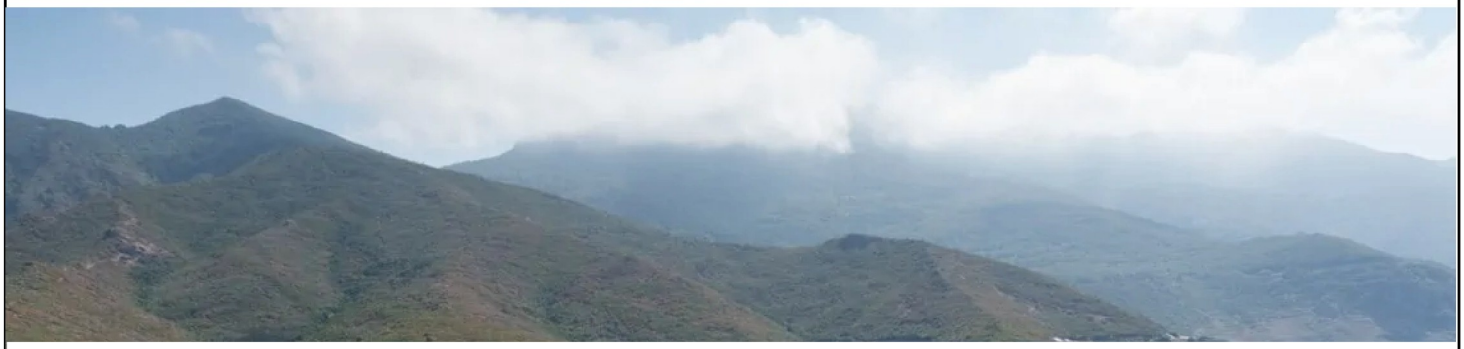
Departement Géographie, aménagement, environnement et développement - Parcours Géomatique et modélisation spatiale (GMS)



ANALYSIS OF WILDFIRES IN HAUTE-CORSE (FRANCE): SPATIAL PATTERNS, SEASONAL DYNAMICS, AND RISK FACTORS



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I. INTRODUCTION

Wildfires represent one of the most significant natural hazards affecting landscapes and societies across France. With approximately **17.5 million hectares of forest** and extensive natural vegetation, the French territory is inherently susceptible to wildfire risk, particularly in Mediterranean and insular regions such as Corsica (Ministère de l'Agriculture, 2023; Ministère de la Transition écologique, 2025). Historically concentrated in the southern part of the country, recent fire seasons have shown a **noticeable expansion of risk both spatially and temporally**, driven by combinations of climatic extremes and human activities (Statistiques & Développement durable, 2024; OFB, 2025). In this context, geomatics offers powerful tools for spatial analysis and risk assessment, enabling the integration of fire occurrence data with environmental and human-related variables.

The **BDIFF (Base de Données sur les Incendies de Forêts en France)** constitutes a key source for analyzing wildfire patterns from 1973 to 2022. Coupled with the DFCI grid and detailed topographic layers from BD Topo, this dataset allows for a **high-resolution spatial examination** of wildfire distribution, seasonal dynamics, density hotspots, and the interaction with landscape features such as vegetation and built-up areas.

II. PROBLEMATIC CONTEXT

Despite ongoing efforts in prevention and firefighting, wildfire risk in France remains persistent and dynamic. In average years, almost **4,000 to 5,400 wildfire events** occur, with annual burned areas that can escalate dramatically during extreme seasons (Ministère de l'Agriculture, 2023; Statistiques & Développement durable, 2024). About **90 % of wildfires are attributed to human causes**, intensifying the importance of spatial patterns related to land use and human presence (Ministère de la Transition écologique, 2025). Furthermore, climate change is projected to **extend the wildfire season, increase fire-prone conditions, and expand the geographic extent of fire risk**, affecting regions historically less exposed (adaptation-changement-climatique.gouv.fr, 2025; OFB, 2025).

In this project, the challenge lies in understanding not only **when and where wildfires occur**, but also how these events relate to environmental features such as vegetation, and the proximity to human settlements. By focusing on the Highlands of Corsica (Haute-Corse), which has been identified as a **fire density hotspot**, this study seeks to disentangle the spatial-temporal dynamics of wildfire recurrence and support more effective land management and fire risk mitigation strategies.

III. METHODOLOGY

This study is based on a geomatics approach combining spatial data analysis and cartographic production to analyze wildfire patterns in France. The methodology follows a structured workflow composed of four main stages: **data collection, data understanding, data analysis, and diagnostic interpretation**.

1. Data Collection

Several authoritative geospatial datasets were used in this study:

- **BDIFF (Base de Données sur les Incendies de Forêts en France)**, managed by the *Institut national de l'information géographique et forestière (IGN)*, which provides detailed information on forest fire occurrences (location, date, burned area, and fire characteristics).
- **DFCI grid (Défense des Forêts Contre l'Incendie)**, a spatial reference grid commonly used for fire prevention and monitoring in France, allowing a standardized spatial analysis of fire distribution.
- **BD TOPO**, also produced by IGN, which supplies detailed topographic and land-use information (transport networks, urban areas, vegetation, hydrography), useful for contextualizing fire events.

All datasets were collected in compatible spatial reference systems to ensure consistency during spatial processing.

2. Data Understanding and Preparation

An exploratory analysis was first conducted to understand the structure, attributes, temporal coverage, and spatial accuracy of each dataset. This step included:

- Verification of data completeness and consistency,
- Selection of relevant variables for the analysis,
- Cleaning and formatting of attribute tables,
- Harmonization of spatial layers to ensure correct overlay operations.

This phase was essential to ensure the reliability of subsequent spatial analyses.

3. Data Analysis and Cross-Referencing

The core of the methodology relies on **data cross-analysis** combining the BDIFF fire records with the DFCI grid and BD TOPO layers. Spatial joins and overlays were performed to:

- Analyze the spatial distribution of fires within the DFCI grid,
- Identify areas with high fire occurrence or recurrence,

- Examine relationships between fire events and territorial features such as land cover, proximity to infrastructures, and urban interfaces.

QGIS was used for all spatial analyses and map production, while **Microsoft Excel** was employed for tabular cross-analyses, statistical summaries, and the generation of graphs supporting the spatial results.

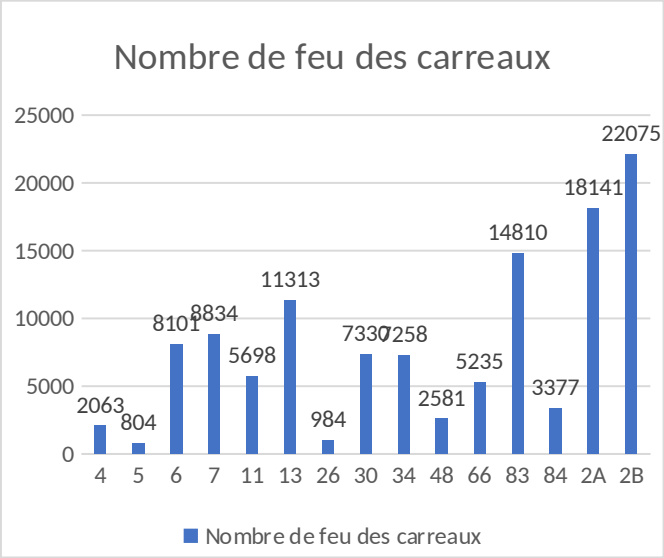
4. Diagnostic and Interpretation

The final stage consists of interpreting the analytical results to establish a diagnostic of wildfire dynamics in France. The spatial patterns observed were analyzed in relation to environmental and territorial factors in order to:

- Identify fire-prone areas,
- Highlight potential risk factors,
- Support a better understanding of wildfire distribution and intensity.

This diagnostic provides a basis for discussing fire prevention strategies and contributes to the broader understanding of wildfire risk management from a geomatics perspective.

IV. SITE SELECTION RATIONALE



The choice of the study area is based on a quantitative cross-analysis of the BDIFF wildfire data and the DFCI grid. By intersecting these datasets, a matrix was created in which **departments are represented in columns, DFCI grid cell identifiers in rows, and values correspond to the number of times each DFCI cell was affected by fires**. This approach makes it possible to calculate, for each department, the cumulative number of DFCI grid cells impacted by wildfires over the study period. The results clearly indicate that **department 2B (Haute-Corse)** records the highest number of DFCI grid cells affected by fires between **1973 and 2022**.

This result reflects not the surface burned at a single time, but rather the **frequency with which fires have impacted one or several DFCI grid cells over time**. In Haute-Corse, a total of **22,075 DFCI grid cells were affected** during the study period, highlighting the recurrent and widespread nature of wildfire activity in this department. This high level of fire occurrence

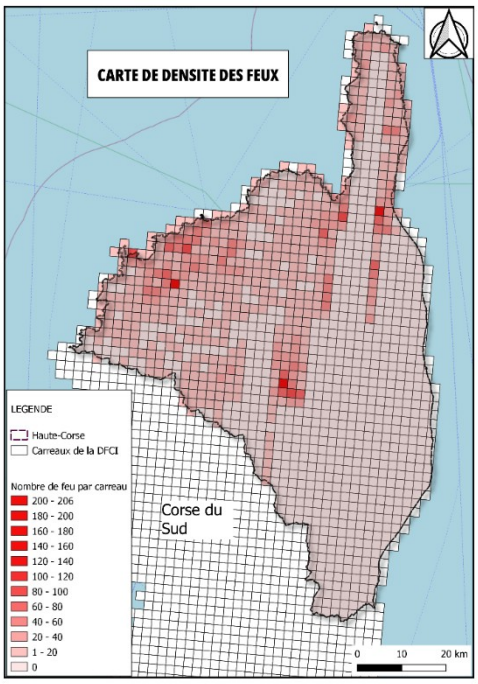
makes Haute-Corse a particularly relevant area for a detailed spatial analysis of wildfire dynamics.

Département	**4**	**5**	**6**	**7**	**11**	**13**	**26**	**30**	**34**	**48**	**66**	**83**	**84**	**2A**	**2B**
Nombre de feu des carreaux	**2063**	**804**	**8101**	**8834**	**5698**	**11313**	**984**	**7330**	**7258**	**2581**	**5235**	**14810**	**3377**	**18141**	**22075**

The selection of department 2B is therefore directly derived from the cross-analysis of the **BDIFF (Base de Données sur les Incendies de Forêts en France)**. This database provides key information on wildfire events, including the year of occurrence, department, and fire alert data. Given its national coverage, the BDIFF dataset allows for a comparative analysis across all French departments. Based on the results of this initial assessment, the dataset was subsequently **filtered to department 2B**, enabling a focused and in-depth analysis of wildfire patterns and spatial characteristics in Haute-Corse.

V. ANALYSIS PRESENTATION

1. Interpretation of the Fire Density Map in Haute-Corse



(Number of fires per DFCI grid cell – global analysis)

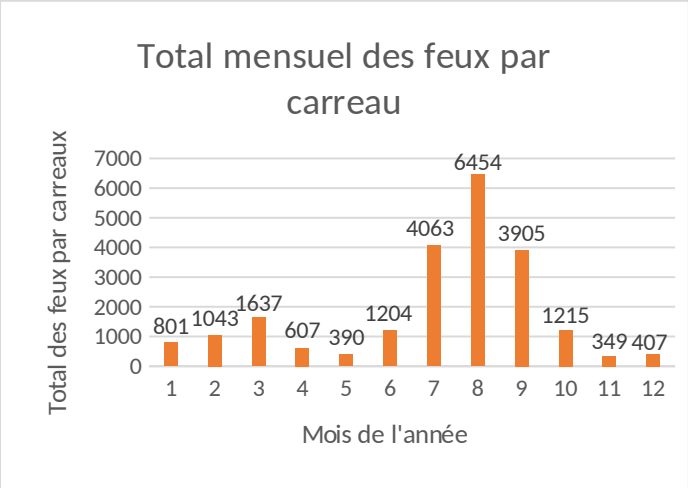
The fire density map represents the **total number of fire events recorded per DFCI grid cell over the entire study period** in Haute-Corse. Unlike the seasonal analysis, this global approach highlights **long-term spatial patterns and fire recurrence**, making it possible to identify areas that are repeatedly affected by wildfires.

The map shows a **heterogeneous spatial distribution** of fire density across the department. While many DFCI grid cells record few or isolated fire events, several zones clearly stand out as **high-density hotspots**, where the number of fires per grid cell is significantly higher. These areas correspond to territories experiencing recurrent fire activity, indicating a persistent vulnerability to wildfires over time.

High fire densities suggest a combination of favorable conditions for ignition and spread, such as vegetation type, topography, climatic exposure, and human influence. In contrast, areas with low fire density appear less frequently affected, either due to more resilient environmental conditions or reduced exposure to ignition sources. This global fire density map therefore provides a synthetic view of **structural fire risk** in Haute-Corse, emphasizing zones where wildfire recurrence is most pronounced.

Overall, this analysis is essential for identifying **priority areas for prevention and management**, as it reveals spatial concentrations of fire activity that may not be apparent in short-term or seasonal analyses. It supports long-term planning strategies by focusing on areas where fire occurrence is not only frequent but also persistent over the years.

2. Results of the Seasonal Wildfire Analysis in Haute-Corse



The seasonal analysis of wildfires in Haute-Corse is based on a cross-referencing of the BDIFF fire records with the DFCI grid. In this analysis, **DFCI grid codes are used as rows, the number of affected grid cells as values, and the months of the year as columns**. This structure makes it possible to quantify the **cumulative number of DFCI grid cells impacted by wildfires for each month**, thereby highlighting seasonal variations in fire occurrence.

The results of this cross-analysis reveal clear **seasonal patterns** in wildfire activity. Certain periods of the year show a significant increase in the number of DFCI grid cells affected by fires, corresponding to months with more favorable conditions for fire ignition and spread, such as higher temperatures, prolonged droughts, and increased human activity. This approach allows for the identification of **peak fire months**, during which wildfire risk is particularly high.

By aggregating fire occurrences at the DFCI grid level, this method emphasizes not only the frequency of fires but also their spatial recurrence throughout the year. The seasonal analysis therefore provides valuable insight into the temporal dynamics of wildfires in Haute-Corse and

supports a better understanding of critical periods for fire prevention, monitoring, and risk management.

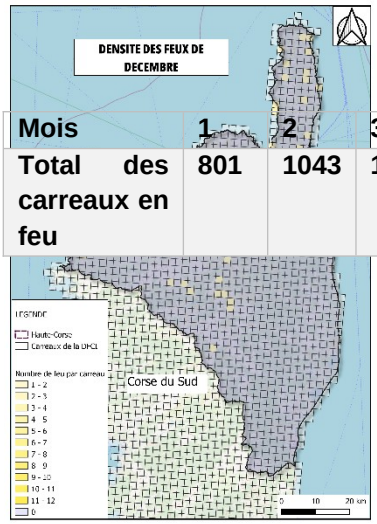
The seasonal analysis highlights a strong **temporal variability** in wildfire occurrence in Haute-Corse, based on the total number of DFCI grid cells affected each month. The results show that wildfire activity is present throughout the year, but with marked seasonal peaks.

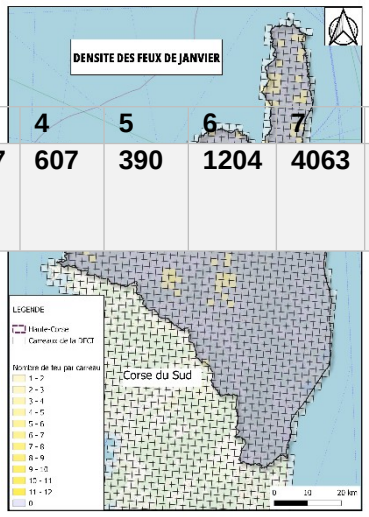
The **summer months** clearly concentrate the highest number of affected grid cells. **August** records the maximum value with **6,454 DFCI grid cells impacted**, followed by **July (4,063)** and **September (3,905)**. This period corresponds to the peak fire season, characterized by high temperatures, dry vegetation, and increased exposure to ignition sources. Together, these three months account for the majority of fire-affected grid cells, underlining the critical importance of summer in wildfire dynamics in Haute-Corse.

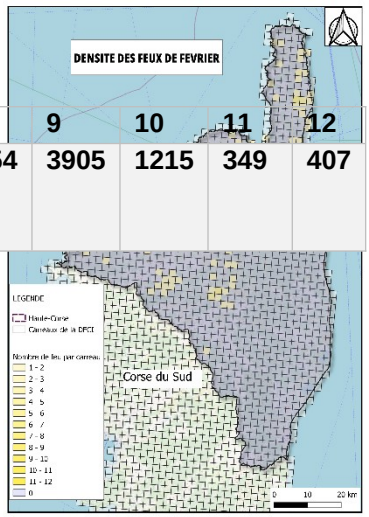
In contrast, **winter and early spring months** show significantly lower fire activity. Months such as **November (349), May (390), and December (407)** register the lowest numbers of affected grid cells. However, the presence of fires even during these periods indicates that wildfire risk in Haute-Corse is not exclusively limited to summer. Overall, this seasonal pattern confirms a strong summer dominance of wildfire activity while also highlighting the need for year-round monitoring and prevention strategies.

- **Winter (December – January – February)**

During winter, wildfire activity remains **relatively limited** compared to other seasons. The maps show a **low density of affected DFCI grid cells**, with fires generally scattered and spatially discontinuous. Fire occurrences during this period are mainly localized and often linked to specific human activities rather than to favorable climatic conditions. Despite the lower number of affected grid cells, the presence of fires indicates that wildfire risk does not completely disappear during winter.



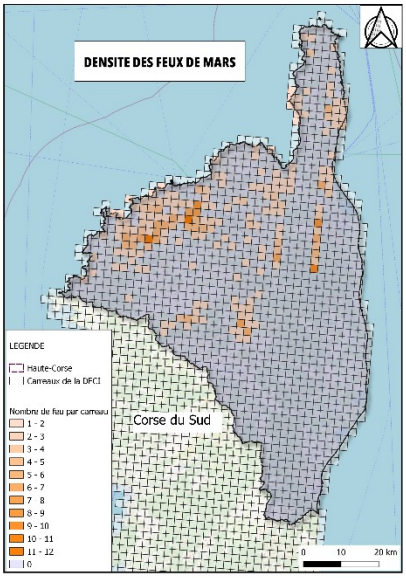


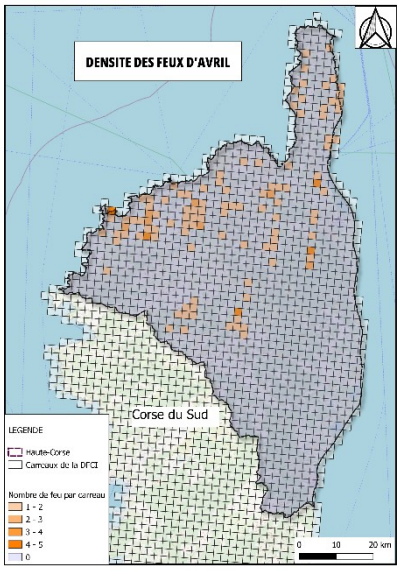


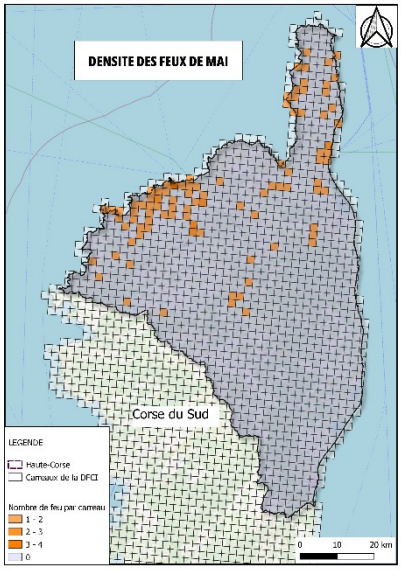
Mois	1	2	3	4	5	6	7	8	9	10	11	12
Total des carreaux en feu	801	1043	1637	607	390	1204	4063	6454	3905	1215	349	407

- **Spring (March – April – May)**

Spring is characterized by a **gradual increase in wildfire activity**, as reflected by the maps. March shows a noticeable rise in the number of affected DFCI grid cells, while April and May remain more moderate. Fire events during this season tend to be more spatially dispersed but begin to affect a wider range of areas. This transition period reflects changing environmental conditions, such as increasing temperatures and vegetation drying, which contribute to a higher fire susceptibility.

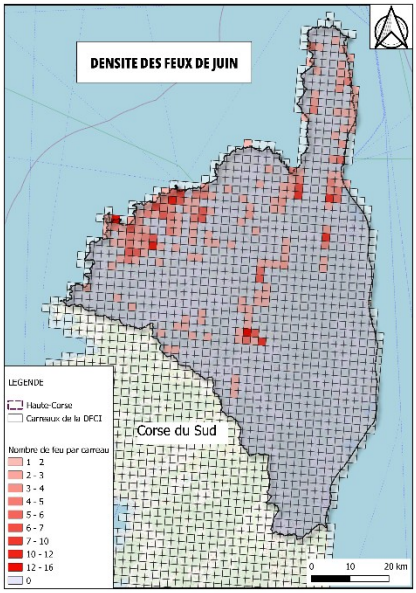


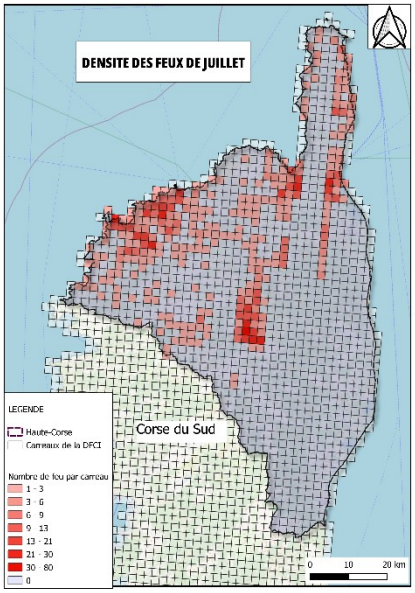


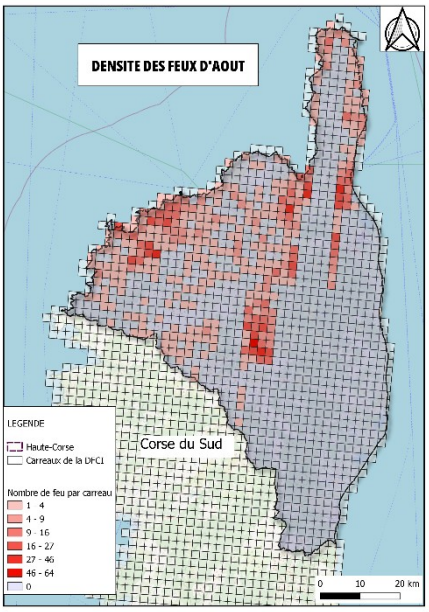


- **Summer (June – July – August)**

Summer clearly stands out as the **most critical fire season** in Haute-Corse. The maps for June, July, and especially August show a **high concentration and wide spatial extent of affected DFCI grid cells**. Fire activity intensifies and becomes more continuous across the territory, indicating both high fire frequency and recurrence. August emerges as the peak month, with the largest number of impacted grid cells, highlighting extreme fire risk conditions driven by prolonged droughts, high temperatures, and strong human pressure.

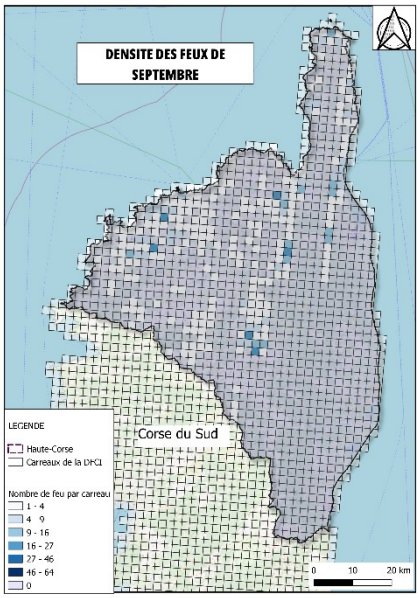


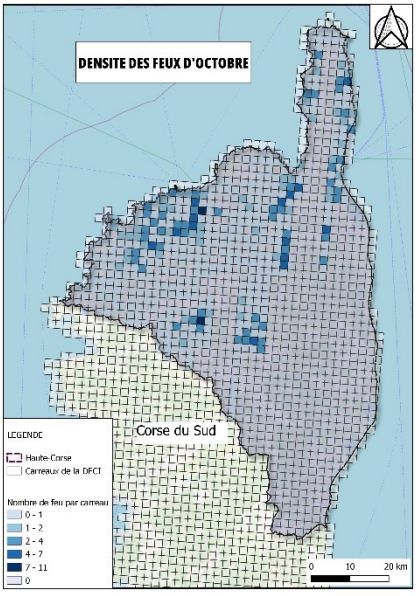


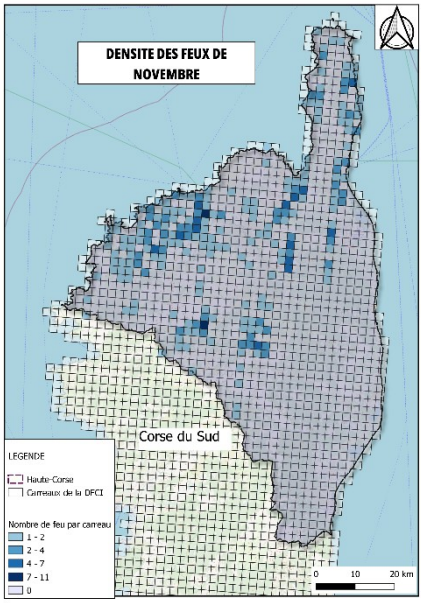


- **Autumn (September – October – November)**

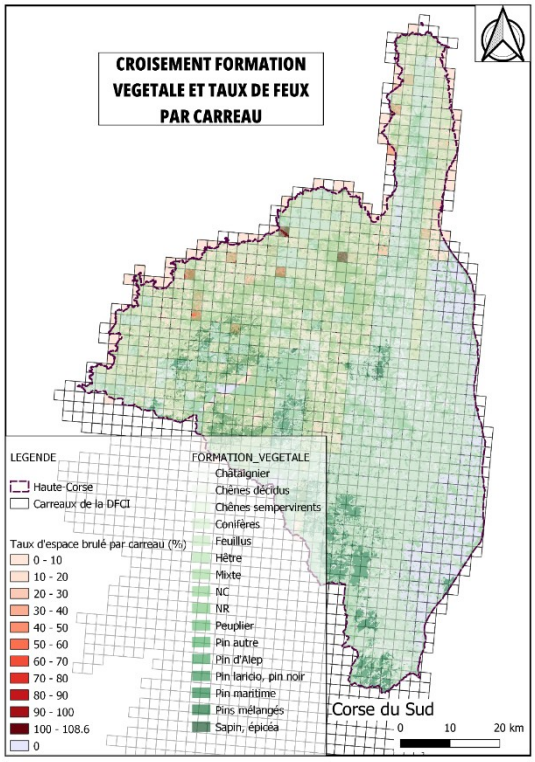
In autumn, wildfire activity begins to **decline progressively**, although September still shows a high number of affected grid cells. The maps illustrate a contraction of fire-affected areas, with fewer grid cells impacted in October and November. This decrease is likely related to cooler temperatures and increased precipitation. Nevertheless, the persistence of fires in early autumn suggests that the end of the summer fire season extends beyond August, requiring continued vigilance during this transitional period.





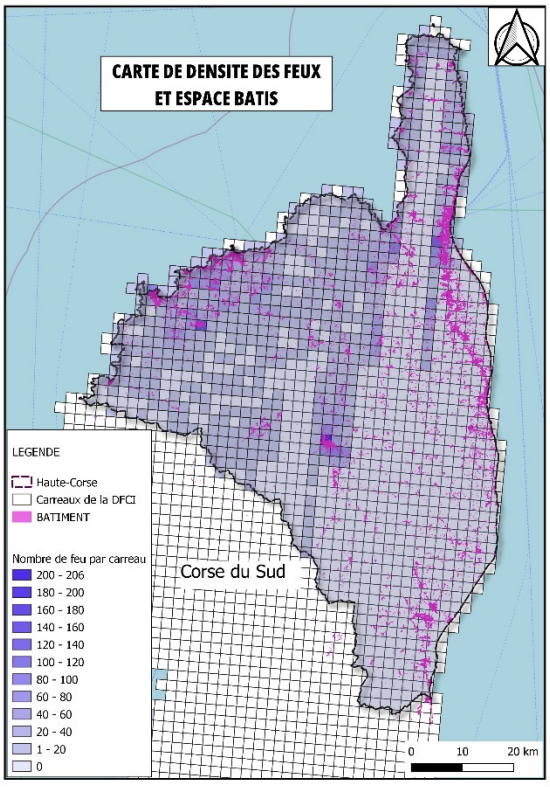


3. *Relationship Between Fire Density and Vegetation Formations in Haute-Corse*



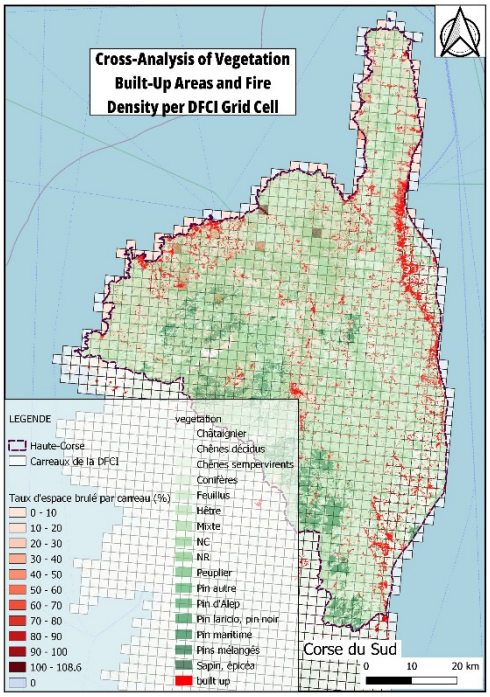
The cross-analysis between the **fire density map** and the **vegetation formation map** highlights a strong relationship between wildfire recurrence and dominant vegetation types in Haute-Corse. Areas with the highest number of fires per DFCI grid cell largely coincide with zones covered by **Mediterranean vegetation formations**, such as shrublands (maquis), garrigue, and open forest stands. These vegetation types are characterized by high flammability, especially during dry periods, which increases both fire ignition probability and fire spread. In contrast, areas dominated by denser forest formations or non-vegetated surfaces generally show lower fire densities. This spatial correspondence confirms the key role of vegetation structure and fuel availability in shaping long-term wildfire patterns and emphasizes the importance of vegetation management in fire-prone areas to reduce recurrent fire risk in Haute-Corse.

4. *Relationship Between Fire Density and Built-Up Areas in Haute-Corse*



The cross-analysis of **fire density and built-up areas** in Haute-Corse shows that areas with high fire recurrence are generally **located outside densely built zones**, confirming that wildfires primarily affect natural and semi-natural landscapes. However, some **high-density fire zones are found near the urban-wildland interface**, indicating a potential risk to human settlements. This pattern highlights the importance of **urban planning and fire prevention measures** around populated areas, such as creating defensible spaces and controlling flammable vegetation, to reduce the impact of recurrent fires on communities.

5. *Cross-Analysis of Fire Density with Vegetation and Built-Up Areas in Haute-Corse*



The cross-analysis of **fire density with vegetation and built-up areas** in Haute-Corse reveals a clear spatial pattern. The highest fire densities are generally located in areas dominated by **flammable vegetation**, such as shrublands (maquis) and open forests, while **built-up areas** show much lower fire recurrence. Fires tend to occur at the **interface between natural vegetation and urban or rural settlements**, highlighting zones where wildfires can threaten human infrastructure. This relationship emphasizes the importance of managing vegetation near populated areas to reduce wildfire risk and protect communities, while also confirming that vegetation type is a primary driver of fire frequency in the region.

VI. GENERAL CONCLUSION

This study provides a comprehensive **geomatics-based analysis of wildfires in Haute-Corse**, integrating spatial datasets from the **BDIFF**, the **DFCI grid**, and **BD TOPO** to examine fire patterns, seasonality, density, and their relationship with environmental and human factors. The results highlight a **strong spatial and temporal variability** in wildfire activity. Summer months, particularly **July and August**, are characterized by the highest fire occurrence, while winter months show limited activity. The fire density analysis identifies **persistent hotspots** with recurrent fires over the period 1973–2022, emphasizing areas of long-term vulnerability.

Cross-analysis with **vegetation formations** demonstrates that highly flammable areas, such as shrublands and open forests, are the primary drivers of fire recurrence. When combined with built-up areas, the analysis highlights **critical interface zones** where wildfires pose a direct risk to human settlements. These findings underline the importance of **targeted fire prevention strategies**, vegetation management, and spatial planning to mitigate wildfire impacts.

Overall, this project illustrates the power of **geomatics and spatial analysis** in understanding wildfire dynamics, identifying risk-prone areas, and supporting effective **fire risk management and decision-making** in Haute-Corse. It also provides a methodological

framework that can be applied to other regions of France or similar Mediterranean environments.

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